1 Impedance

Veličina v obvodech střídavého proudu (téměř bez výhrady sinusový průběh).

Uvažujeme ji proto, že se stoupající frekvencí kapacitou/indukčností dochází na určitých prvcích (reaktančních) k fázovému posunu. To je možné zobrazit v komplexní rovině.

Do Gaussovy roviny vynášíme poměrnou impedanci.

2 Obecně

$$\hat{Z} = R \parallel \hat{X} \rightarrow \frac{\hat{Z}}{R} = \frac{R \parallel \hat{X}}{R} \qquad \omega = 2\pi \cdot f \qquad \varphi = \arctan \frac{\hat{Z}}{R} \qquad \hat{Z}_{dB} = 20 \cdot \log \left| \frac{\hat{Z}}{R} \right|$$

3 dB

DeciBell – bezrozměrná jednotka přenosu

$$1B = \log \frac{P_2}{P_1} \rightarrow 1dB = 10 \cdot \log \frac{P_2}{P_1}$$

Protože je 1 B moc velká jednotka, používá se dB. Proto 10x. Při použití U/I 20x viz:

$$P = R \cdot I^2 = \frac{U^2}{R} \rightarrow 1dB = 20 \cdot \log \frac{U_2}{U_1} \left| \frac{I_2}{I_1} \right.$$

4 Tau

Vychází z přechodových dějů $\tau = R \cdot C \qquad \tau = \frac{L}{R}$

Časová konstanta

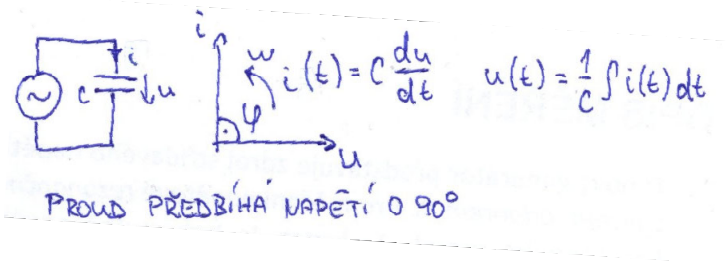
5 Mezní frekvence f_o / ω_0

Frekvence zlomu charakteristiky

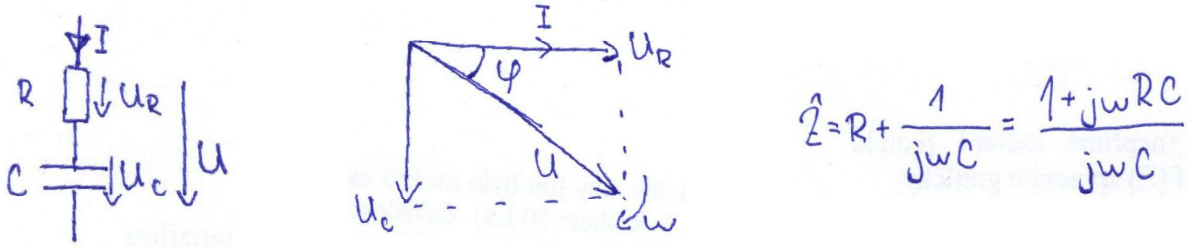
$\omega/\omega_0 = 1$

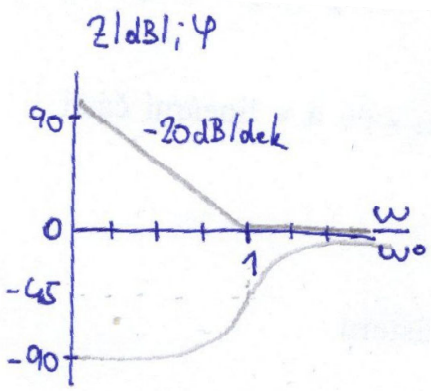
V tomto bodě je $\mathbf{Z}_{dB} = \pm 3$ dB (kolem bodu je zaoblené, pro zjednodušení se kreslí ostrý zlom)

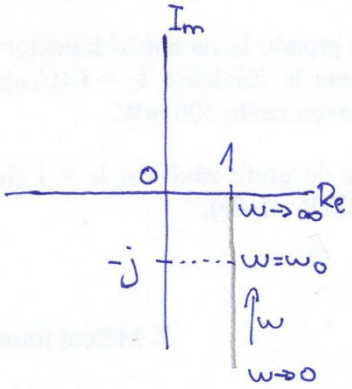
6 Kondenzátor



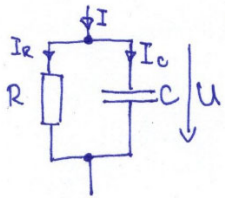
6.1 RC sériově

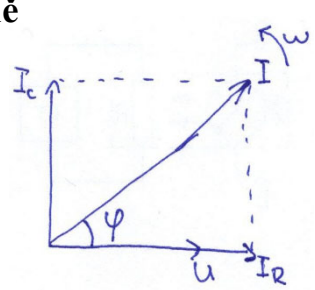




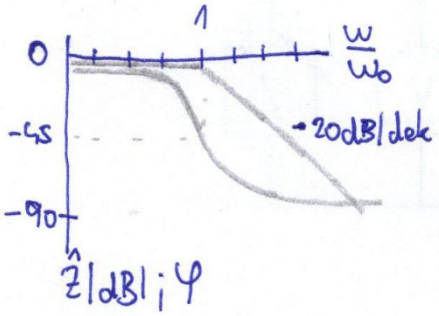


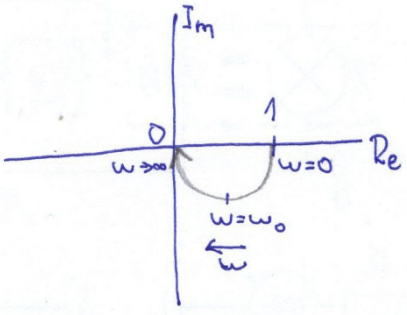
6.2 RC paralelně



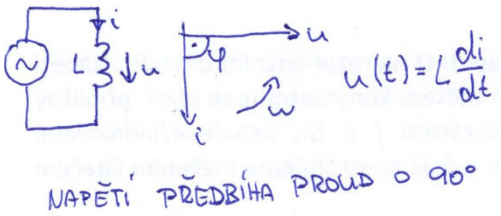


$$\hat{Z} = \frac{R\cdot\frac{1}{j\omega C}}{R+\frac{1}{j\omega C}} = \frac{R}{1+j\omega RC}$$



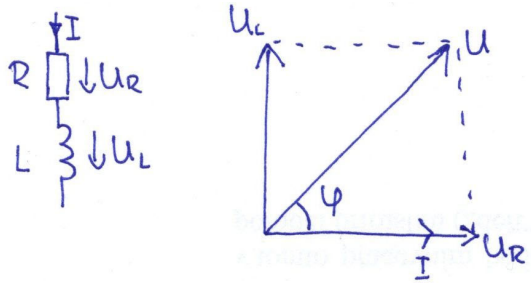


7 Cívka

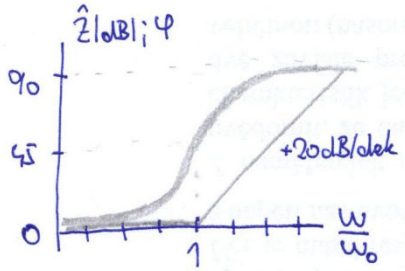


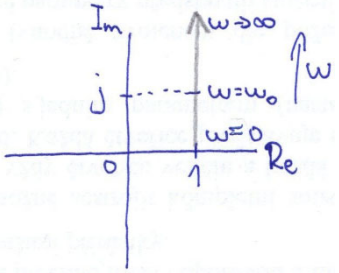
$$X_L = j\omega L$$

7.1 RL sériově

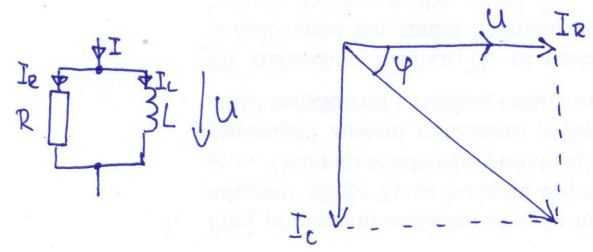


$$\hat{Z} = R + j\omega L$$

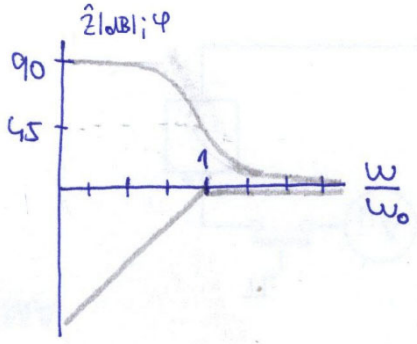


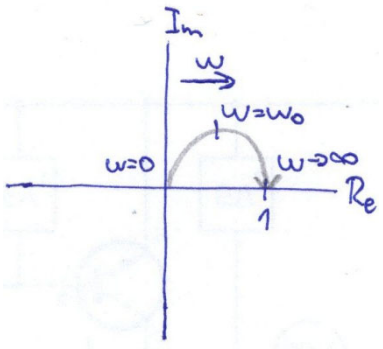


7.2 RL paralelně



$$\hat{Z} = \frac{R \cdot j\omega L}{R + j\omega L}$$





8 Rezonanční obvody

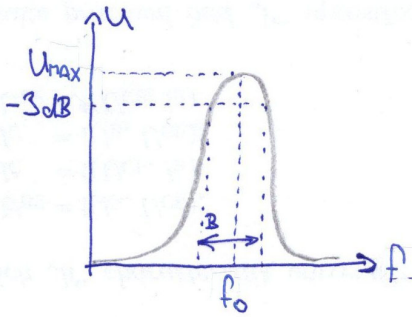
Rezonanční frekvence – Thomsonův vztah $\omega_0 = \frac{1}{\sqrt{LC}}$

Poměrné rozladění $F = \frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}$

$\frac{\hat{Z}}{R} = 1 + j Q F$

8.1 Šíře pásma a jakost

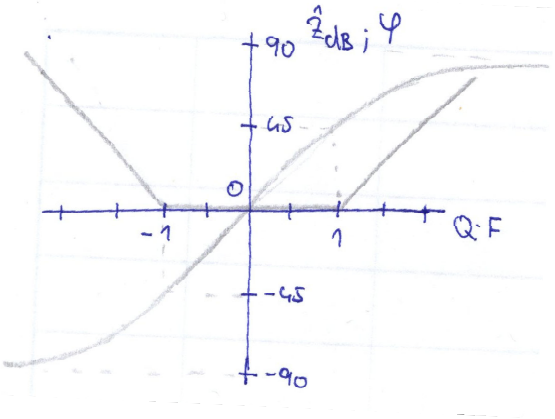
Šíře pásma = rozsah frekvencí mezi poklesem o 3 dB.

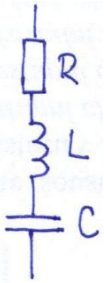


$Q = \frac{f_0}{B}$

$Q = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 RC}$

8.2 Sériový





8.3 Paralelní

